

Execution and transaction costs

How a flat cost and full fills were replaced with a standard cost model, and what that model says about cost as the account grows.

HEADLINE FIGURES: HISTORICAL BACKTEST · SELECTED CONFIGURATION. NO LIVE OR PAPER-TRADING RESULTS ARE SHOWN.

about 40 bp Modeled cost per round trip, in basis points, at a \$1M start	30 to 55 bp The same cost, approximately, from a \$100k start to a \$5M start	3.04× Simulated ending multiple after modeled costs, \$1M start, about two years	None Figures measured on the strategy's own orders
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Selected configuration result. Configuration selection and evaluation overlap in this research window, so it is not independent out-of-sample evidence. Deflated Sharpe ratio: 0.85.

The problem

The strategy is a long-only, market-hours systematic equity strategy that trades short-horizon relative-value dislocations between single stocks and statistically formed peer groups, with a portfolio-level risk governor. My first reported backtest of it assumed that every order filled in full at the signal bar's price, for a small flat fee per round trip.

Real orders cross part of the bid-ask spread, push the price against themselves, and sometimes cannot be completed. A flat cost with full fills hides all three. It makes cost look the same at every account size, so it says nothing about capacity, and it flatters thinly traded names most.

Paper trading cannot close the gap. A paper account simulates its fills without sending orders to the market, so it cannot calibrate a cost model at any size.

The model

Every part of the model is textbook. Each side of a trade pays half of an assumed bid-ask spread plus a square-root market-impact term.

$$\text{cost per side} = s/2 + Y \cdot \sigma \cdot \sqrt{(Q/V)}$$

Here s is the spread assumed for the stock's liquidity tier, σ its daily volatility, Q the order size, V its daily dollar volume and Y the impact coefficient. Three rules sit around the formula.

- **Delay.** Most orders start at the bar after the one that produced the signal.
- **Participation cap.** An entry is worked, meaning traded gradually over a window, and may take only a capped share of the volume that trades in that window. An entry may fill only in part, with the remainder dropped, and the model declines some entries outright. Exits are not capped: every exit is assumed to complete, at the modeled price and cost.

- **Implementation shortfall.** The gap between frictionless profit (every share at the decision price, no cost) and executable profit is reported by component: delay, spread, impact, fees and unfilled shares. In the shortfall study the identity closed to the dollar on every row.

The impact coefficient

I had Y estimated two ways from public trade-and-volume data. The two disagree: in most of the universe one reads materially higher than the other. I carry that gap as uncertainty.

Each estimate, combined with a prior from the academic literature, gives a distribution for Y. The model's fixed value is conservative under the lower-reading estimate and roughly central under the higher-reading one, and neither measures my own orders. I do not disclose the value, the cap or the spread tiers.

What the studies found

More than a dozen studies ran on a one-minute fill simulator, over the trade record of the first reported backtest, which is superseded. A finding counted only if it held in at least three of four half-years and on both ordinary and unusually busy days, tuned on the first half of the sample and reported on the second.

1. **At small size, participation decides fills, not cost.** In a four-level stress of the cap at a \$1M account size, the share of frictionless profit kept was indistinguishable across the three higher levels. The gap at the lowest level was unresolved at \$1M and resolved, narrowly, at three times the size. Lowering the cap removes fills faster than it removes impact.
2. **A better exit rule improved completion. I claim no price gain.** Against the first version of the worked-exit rule, a refined version raised the filled share of exit orders under every fill assumption tested. A gain of about 2 to 3 basis points (bp) per order appeared under the most favorable fill assumption and disappeared under a stricter one. An earlier, more flattering reading was an artifact of the reference price, and I corrected it.
3. **An optimal-control formulation added nothing to a simple rule.** The problem had square-root temporary impact, a participation constraint and a penalty for unfilled shares. The first-order condition favored trading at the cap at the start of every entry, but the full solution still spread most orders across its horizon. On data held out from tuning, that trailed the best simple schedule, with a bootstrap interval that excludes zero.
4. **A learned execution-timing policy did not beat a fixed schedule.** Across more than two hundred configurations, its gain was matched by a copy trained on scrambled inputs and exceeded on random windows with no signal in them. One cell favored a learner on uncorrected intervals, and a correction for the number of configurations would remove it.
5. **Capacity is not one number.** The binding constraint differs across segments of the universe: in some the order outgrows the volume available, in others modeled cost binds. On the pooled sample the order in which segments saturate did not depend on the participation assumption. Between half-years only the first segment to saturate repeated, in three of four and on few trades, and the levels moved by large factors, so I report no capacity figure.
6. **From frictionless to executable profit.** At \$1M a worked, participation-capped schedule kept most of the frictionless profit. A stand-in for a single unworked order kept clearly less, mostly through unfilled shares. The worked schedule led in all four half-years, but a small share of trades carries its dollar lead,

and the lead narrows under a less favorable fill-price convention. Its t-statistic is about 2.1 at \$1M and 2.2 at three times the size, with standard errors that allow for dependence within sessions and within peer groups. A coarser grouping, unreliable with so few groups, gives 1.5 and 1.6, which is not significant. On unusually busy days the two are a statistical tie, so the lead is not established there.

Cost and account size

The cost model is now part of the walk-forward replay, the historical backtest. I had the selected configuration, the one behind my headline figures, run through it at three starting balances. Each row is the same simulation with order sizes scaled to the stated balance. Small-account frictions such as minimum fees are not modeled.

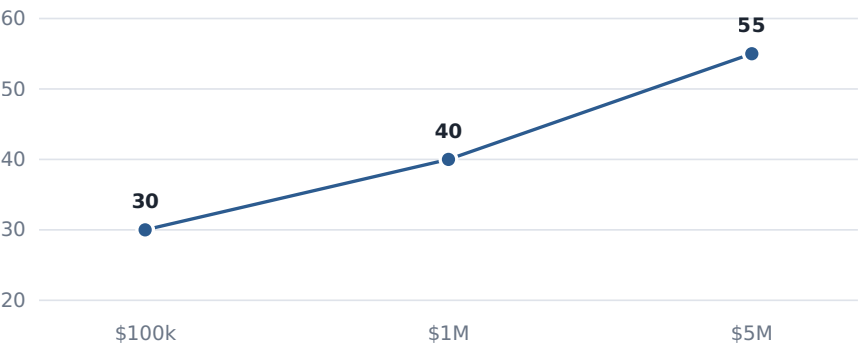
STARTING BALANCE	COST PER ROUND TRIP	ENDING MULTIPLE
\$100k	about 30 bp	higher
\$1M	about 40 bp	3.04×
\$5M	about 55 bp	lower

Historical backtest · selected configuration. Point-in-time universe, modeled transaction costs, profits reinvested, Jun 2024 – Jul 2026. Cost rounded to the nearest 5 bp.

Selected configuration result. Configuration selection and evaluation overlap in this research window, so it is not independent out-of-sample evidence. Deflated Sharpe ratio: 0.85.

Cost per round trip is total modeled cost over total entry notional. It includes the original flat fee, which is kept, both half-spreads, impact on both sides and the delay to the next bar. Unfilled shares and declined entries carry no charge. They show up as fewer and smaller trades.

The half-spread charge is flat across balances, and the impact term grows with order size.



Basis points per round trip, rounded to the nearest five, by starting balance

Modeled cost per round trip by simulated starting balance. Historical backtest · selected configuration, order sizes scaled to the balance.

At the other two balances the table gives a direction only for the ending multiple. The cost column is the reliable read.

What is not modeled

- **Own orders.** Y is calibrated to public tape data and has not been calibrated on the strategy's own orders.
- **Financing.** The simulated book can use leverage, and no financing cost is charged.
- **Incomplete exits.** The replay assumes every exit completes. In the one-minute studies, exits at larger sizes did not always complete inside their working window.
- **Permanent impact.** Impact is a temporary charge on each order. My fills do not move the prices that later fills and marks see.
- **Intraday volume persistence.** The optimal-control solver treated each minute's volume as independent. Real volume persists, so its expected fill for a spread schedule is optimistic.
- **Queue position, latency and crowding.** None is modeled. The capacity projections hold profit per trade fixed.

Limits

- **Projection, not measurement.** No figure in this brief is measured on the strategy's own orders.
- **Where the studies ran.** The six findings come from the superseded trade record, whose list of tradable names contained hindsight. I have not repeated the studies on the selected configuration.
- **Busy days.** The busy-day half of the repeatability rule rests on a small number of sessions, so a pass or a fail there is weak evidence.
- **One implementation.** The fill model has no second-implementation check and no parity run against the trading engine. The worked-order methods have not been checked against real fills.
- **Fill-price convention.** The replay fills at a bar-average price, a proxy for the volume-weighted price that carries some error. In an earlier replay on the superseded universe, an adverse bound on every fill removed essentially all of the simulated gain. That is a bound, not a forecast, and I have not re-run it on the selected configuration.
- **The headline backtest.** The 3.04× is one implementation of one two-year window. Its minimum-liquidity requirement was chosen after testing five settings on that window. Its deflated Sharpe ratio, a probability and not a reduced Sharpe, is 0.85 against the 240 full-window simulations on file. That count is simulations on file, not ideas tried, and may understate the search. A small fraction of its trades accounts for more than half of net profit. A compounded two-year multiple moves by about $\pm 0.5\times$ when a run changes slightly.
- **Who did what.** Implementation uses extensive AI-assisted development; the questions, the standard and the adoption decisions are mine.

Selected material presented for educational and professional purposes. Proprietary signal definitions, parameters, portfolio construction rules and execution logic are intentionally omitted. All performance figures are historical backtests unless labeled otherwise; no live or paper-trading results are shown. Historical and simulated results do not guarantee future performance. Nothing here is an offer or investment advice.